PROJECT REPORT

DATABASE MANAGEMENT SYSTEM

(Hospital Management System)



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INTRODUCTION

The Hospital Management System uses 10 tables to store and manage hospital information efficiently.

Department: Stores department ID, department name, and location.

Doctor: Stores doctor ID, name, specialization, phone number, department ID, and consultation fee.

Patient: Stores patient ID, name, age, gender, phone number, and address.

Nurse: Stores nurse ID, name, phone number, and department ID.

Room: Stores room ID, room type, room charges, and room status.

Appointment: Stores appointment ID, patient ID, doctor ID, appointment date, time, and status.

Admission: Stores admission ID, patient ID, room ID, admission date, and discharge date.

Treatment: Stores treatment ID, patient ID, doctor ID, treatment description, and treatment date.

Medicine: Stores medicine ID, medicine name, price, and stock quantity.

Billing: Stores bill ID, patient ID, admission ID, bill date, total amount, and payment status.

DATABASE CONCEPT USED

The project uses the following database concepts:

Primary Key: Uniquely identifies each record in a table.

Foreign Key: Establishes relationships between related tables.

Constraints: Maintain data accuracy and integrity.

SQL Queries: Used to insert, retrieve, and manage data.

Joins: Combine information from multiple related tables.

Aggregate Functions: Functions such as COUNT () and SUM () help calculate totals and generate summaries.

SQL IMPLEMENTATION

The system is implemented using MySQL. SQL commands are used to create the database, define tables, insert sample records, and retrieve information.

The implementation includes:

Creating the HospitalDB database.

Creating 10 tables.

Defining primary and foreign keys.

Inserting sample hospital records.

Displaying patient and doctor information.

Retrieving appointment and admission details.

Generating billing summaries.

Checking room availability.

Note: Insert the complete SQL code in this section, using a readable font.

OUTPUT AND RESULTS

The SQL queries were executed to verify the working of the Hospital Management System.

The following outputs demonstrate the successful implementation of the database:

List of all 10 database tables.

Patient records.

Doctor records.

Appointment details.

Patient and doctor information using JOIN.

Department-wise doctor count.

Available hospital rooms.

Billing records and unpaid bills.

ADVANTAGES

Maintains organized hospital records.

Reduces manual paperwork.

Makes patient information easier to retrieve.

Helps manage appointments and admissions.

Maintains doctor and nurse information.

Supports room and billing management.

Uses primary and foreign keys to maintain data integrity.

LIMITATIONS

The project uses sample data for demonstration.

It does not include a graphical user interface.

User authentication and access control are not implemented.

Online appointment booking is not available.

Automated billing and payment processing are not implemented.

CONCLUSION

The Hospital Management System demonstrates how a relational database can be used to manage hospital information efficiently. The project consists of 10 tables connected through primary and foreign keys wherever required. MySQL queries are used to store, retrieve, and manage information about patients, doctors, nurses, appointments, admissions, treatments, rooms, medicines, and billing.

This project helped in understanding important DBMS concepts, including table creation, data integrity, relationships, SQL queries, joins, and aggregate functions. The system provides a basic foundation for developing a more advanced hospital management application.

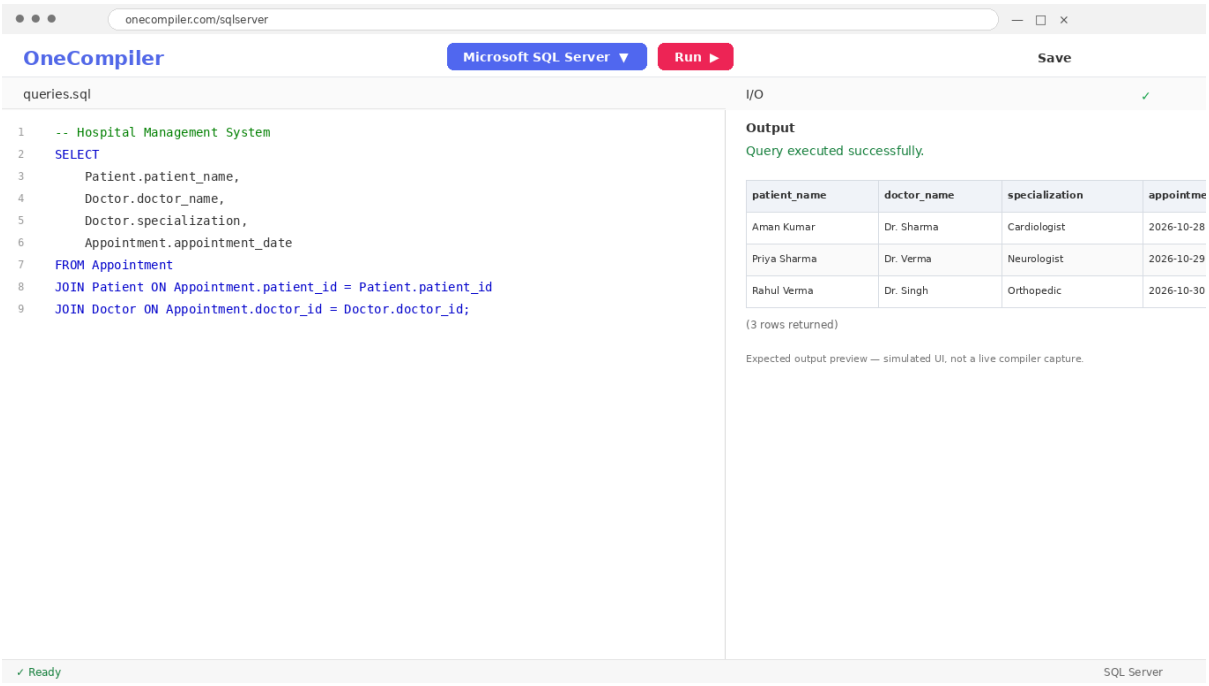
REFERENCES

MySQL Official Documentation — https://dev.mysql.com/doc/

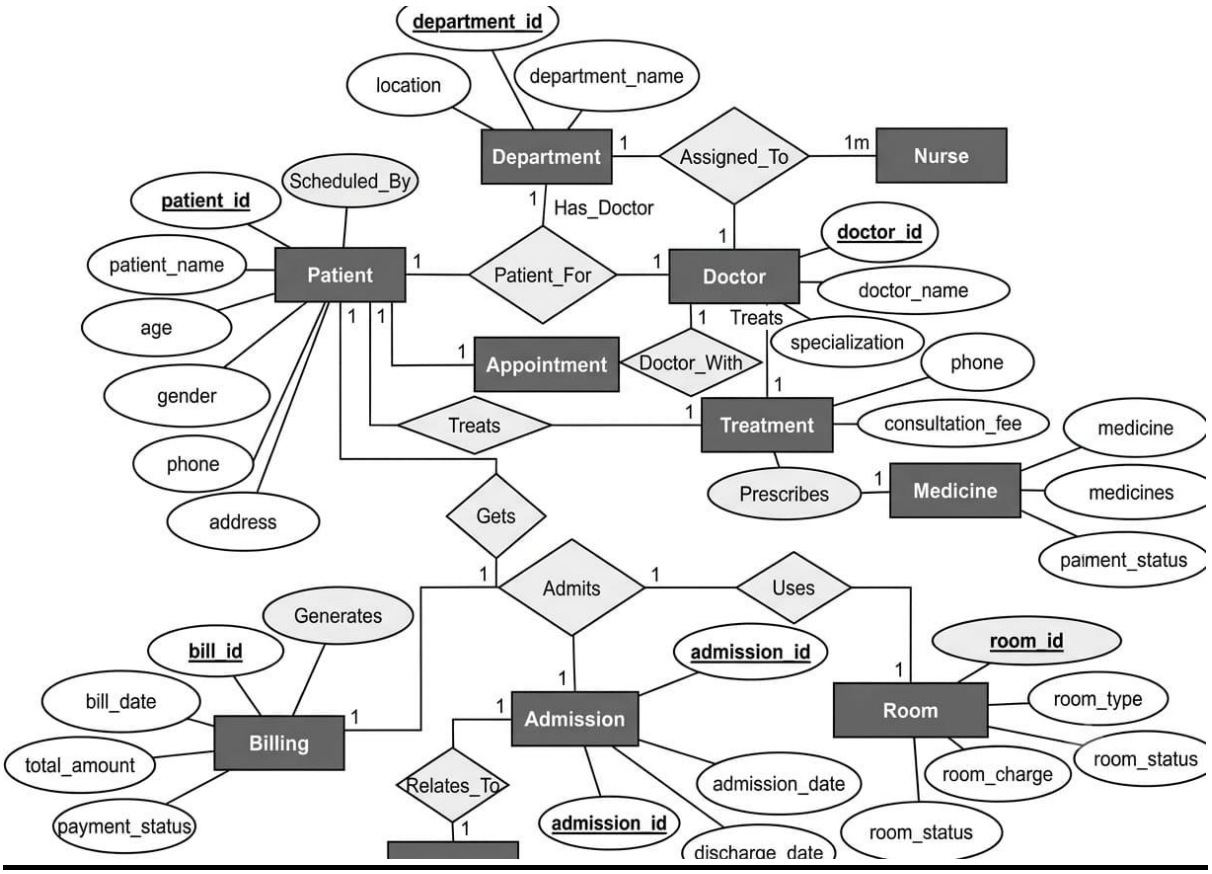
DBMS classroom notes and study material.

SQL concepts and examples discussed during practical sessions.

CODE OUTPUT OF PROJECT



ER DIAGRAM


department_id
location
department_name
Department
Assigned_To
Nurse
Scheduled_By
patient_id
Has_Doctor
patient_name
Patient
Patient_For
Doctor
doctor_id
doctor_name
age
Treats
specialization
Appointment
Doctor_With
phone
gender
Treats
Treatment
consultation_fee
medicine
phone
Prescribes
Medicine
medicines
address
Gets
paiment_status
Admits
Uses
Generates
bill_id
admission_id
room_id
bill_date
Admission
Room
room_type
Billing
total_amount
Relates_To
admission_date
room_charge
room_status
payment_status
admission_id
discharge_date
room_status